Woven Trust: Consensus and Governance Anchored in Verifiable Reputation

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Abstract

We present Woven Trust (*Confianza Tejida*), the architecture of a decentralized network in which authority — eligibility for the finality committee, dispute juries, and governance — is derived from a reputation metric computed deterministically over evidence recorded on chain, rather than from computational work or capital at stake. Reputation is computed per domain from two objective sources: bilateral attestations of settled transactions and a graph of vouch bonds, propagated by an algorithm of the EigenTrust family in fixed-point (i128) arithmetic, so that every node arrives at bit-identical values. Powers that require cross-domain integrity use a concave aggregate of the four domains that vetoes profiles lacking minimum coverage. Sybil and collusion resistance is achieved through defense in depth: graph-proximity discounting, dense-community damping, vouch bonds with probation windows, and a weighted-sortition judicial layer. Finality is granted by a committee rotated per epoch, drawn by sortition seeded from a grinding-hardened commit-reveal beacon, and protected by an entrenchment guard that freezes finality — reversibly — if any single entity's weight crosses a ceiling. We document the dual-asset economic model with issuance tied to measured service, the bootstrap procedure with protocol-forced founder dilution, and an adversarial evaluation backed by a public validation ledger: 17/17 attack scenarios resisted by the reputation engine, two-hop collusion jury capture quantified and closed, and consensus survival across validator dropout and rejoin at epoch boundaries in two independent test environments. We close with the open limitations: proof of unique personhood, transport metadata, and parameters still under calibration.

1. Introduction

Open consensus systems anchor their security in a resource the adversary must acquire. Proof of work anchors it in energy [8]; proof of stake, in capital. Both resources are market-purchasable, so network security is, in the limit, a function of the attacker's budget. This work explores a different anchor: **reputation-time** — verifiable behavioral history within the system itself, which accrues slowly, decays with disuse, and cannot be transferred. An adversary cannot rent years of history attested by independent counterparties; it must produce that history under the same rules that watch every other participant.

The classical difficulty with reputation metrics is twofold. First, the Sybil attack [2]: if identity creation is free, any aggregable metric can be inflated with fabricated identities. Second, collusion: coordinated subsets that attest to one another can manufacture signal without any real economic activity. A third problem, less treated in the literature, is institutional: in platform reputation systems [7], the operator custodies the database and can edit it, so the metric inherits the trust — and the corruptibility — of its custodian.

Contributions. This paper describes a complete system, implemented and adversarially validated, with the following contributions:

1. A **multi-domain reputation scheme** computed exclusively from objective on-chain evidence (bilateral attestations of settled transactions; vouch bonds), with no subjective scoring, per-domain temporal decay, and anti-specialist concave aggregation (§4).
2. An **anti-collusion battery in depth** — proximity discounting, community damping, probation-windowed bonds, and a capped liability cascade — evaluated against a catalogue of attacks (§5, §6).
3. A **reputational-committee finality mechanism**: weighted sortition over a commit-reveal beacon with a reputable-weight guard (anti-grinding and anti-freeze), domain-separated signed votes, and an entrenchment guard with reversible halt (§7).
4. A **dual-asset economic model** in which issuance pays for algorithmically measured service and no discretionary authority mints or allocates (§9).
5. A **validation methodology** of evidence-over-claim, with a public ledger tying every declared property to a reproducible run and the hash of the binary that produced it (§11).

The system is implemented on top of a general-purpose blockchain runtime; every computation path relevant to consensus runs in fixed-point integer arithmetic to guarantee determinism across architectures.

2. Related Work

Trust propagation. EigenTrust [1] computes a global trust value as the fixed point of a Markov chain over the interaction graph, anchored by a pre-trust vector. Our engine belongs to this family (§4.3), with two differences: pre-trust is derived from objective evidence rather than from an a priori trusted set, and the computation runs per domain in deterministic fixed-point arithmetic. The teleportation structure is analogous to PageRank's [6].

Structural Sybil defenses. SybilGuard [3] and its successors exploit the fact that fabricated identities connect to the honest graph through few cuts. Our proximity discount and dense-community damping (§5) follow the same structural principle, applied to the weight of evidence rather than to identity admission.

Committee finality. The separation between block production and committee-voted finality parallels BFT-style finality gadgets (e.g., GRANDPA [9]). The central difference is the source of voting weight: non-transferable reputation rather than economic stake on deposit.

Verifiable delay functions. The proposed hardening of the beacon against last-revealer bias uses VDFs [4].

Platform reputation. Centralized review systems [7] aggregate subjective opinions under an editable custodian; the present design inverts all three decisions (facts instead of opinions, multi-domain instead of scalar, universal recomputation instead of custody).

Institutional precedents. The viability of exchange orders sustained by reputation without state enforcement is historically documented; see Milgrom, North, and Weingast's analysis of medieval mercantile law [5] and, on self-governance of common-pool resources, Ostrom [10].

3. System Model and Threat Model

3.1 Actors and Assumptions

Participants are identified by cryptographic pseudonyms: sr25519 keypairs; the address derived from the public key is the member's complete identity within the system. There is no link to any legal identity, by design (the network layer is discussed in §12). Nodes replicate the chain and deterministically recompute the entire reputation state; no node holds privileged state.

We assume the usual partially synchronous network model, standard cryptography (sr25519 signatures, collision-resistant hashes), and an adaptive adversary who can: (a) create unlimited identities at negligible marginal cost, (b) coordinate subsets of owned or bribed identities, (c) withhold or reorder its own protocol messages, and (d) participate honestly for extended periods before deviating.

3.2 Security Objectives

- **S1 (anti-Sybil):** identities without independent evidence accrue no consensus weight, regardless of their number.
- **S2 (anti-collusion):** the weight obtainable by a coordinated subset grows with its evidence external to the subset, not with its internal activity.
- **S3 (non-transferability):** weight cannot be bought, sold, or inherited; it decays without activity.
- **S4 (determinism):** every node computes bit-identical weights from the same chain prefix.
- **S5 (finality safety):** no block is finalized without a supermajority of the reputational weight of the sitting committee; under excessive weight concentration, the system prefers to halt finality (reversibly) over finalizing under capture.
- **S6 (non-discretionarity):** no account holds discretionary interpretation, editing, or allocation power; wherever human judgment is required, it is resolved by sortition (§8).

3.3 Considered Attack Catalogue

Sybil rings; collusion clusters (mutual admiration); whitewashing (exit and re-entry); slashing targeted at one of the victim's domains; vouch-kamikaze and vouch-bomb attacks (§6); beacon grinding and reveal withholding (§7.3); finality freeze by a zero-weight participant (§7.3); two-hop collusion jury capture (§8); committee capture during the bootstrap window (§10).

4. The Reputation System

4.1 Admissible Evidence

Only two types of events generate reputation signal, both objective and verifiable on chain:

1. **Bilateral attestation.** The two counterparties to an on-chain settled transaction record the fact (the transaction closed; what was committed was delivered). There are no graded scores or third-party reviews: only genuine counterparties to a real transaction can attest, which imposes an economic cost on signal fabrication and eliminates review-bombing by construction.
2. **Vouch bond** (§4.2).

Which event types are admissible is fixed by public, recomputable rules, with no discretionary admission authority (objective S6). The bootstrap key that seeds the initial event types expires automatically at a fixed block height (§10).

4.2 The Vouch as a Bond

A member may vouch for another's entry. The vouch: (i) locks a fraction of the sponsor's reputation, in the same domain, as a pledge; (ii) the pledge is released gradually as the sponsored member accrues **independent** reputation, defined as evidence from counterparties whose graph proximity to the sponsor falls below a threshold; (iii) the vouch transfers no weight: a member whose only edge is its sponsor has zero structural weight; (iv) fraud by the sponsored member propagates loss up the vouch chain, capped per incident (§6). The vouch structure forms a directed graph rooted at genesis, public and analyzable: Sybil rings present detectable topological signatures.

4.3 Propagation Engine

Let G_d be the evidence graph of domain d (edges weighted by attestations and vouches, with the discounts of §5). The weight s_d of each identity in domain d is computed as the fixed point of a propagation iteration of the EigenTrust family [1]:

- **Pre-trust:** the anchor vector is not an a priori trusted set but the distribution of objective evidence (settled transactions, verified work) held by each identity.
- **Teleportation:** with fixed probability, the iteration jumps to the pre-trust distribution, which bounds the influence obtainable by dense subgraphs lacking external evidence (analogous to PageRank's damping factor [6]).
- **Deterministic fixed point:** all arithmetic is integer (i128, fixed scale). No floating point appears on any consensus path; the result is bit-identical across every architecture (objective S4).

4.4 Four Domains and Concave Aggregation

Reputation is computed in four independent domains — exchange, work, judgment, and governance (in the project's nomenclature, the four suits of the deck) — and each domain admits only evidence of its own kind. For powers that require cross-domain integrity (finality committee, juries, vouching authority), the four values are aggregated as:

$$\mathrm{agg}(s) = (1-\lambda)\cdot \min_d(s_d) + \lambda \cdot \mathrm{med}_d(s_d)$$

with $\lambda \in [0,1]$ a blending dial (provisionally $\lambda = 0.5$; an open parameter in final calibration, §12). The concave form preserves the anti-specialist property — covering fewer than half of the domains yields a zero median and a near-zero aggregate for every λ, so λ **cannot** rescue an under-covered profile — while it softens the pure minimum, which adversarial analysis showed to be weaponizable: with agg = min, a strike targeted at a single domain zeroes out the victim ($-100\,\%$); with $\lambda = 0.5$ the wound is bounded to $\approx -50\,\%$ in the ideal symmetric case ($-47\,\%$ measured on real profiles on hardware; ledger, row 16). The value $\lambda = 0.5$ is provisional and pending re-validation against the full attack battery (§12).

4.5 Decay

Weight decays multiplicatively per epoch in the absence of new evidence, per domain, with a half-life of $\approx$ 24 months (per-epoch retention $\rho = 0.999052$). Decay implements temporal non-transferability (S3): there are no aged accounts with resale value, and the advantage of early participants dissolves without intervention (§10).

5. Anti-Collusion Defenses

No single defense satisfies S2 in isolation; the system stacks four, all deterministic:

1. **Proximity discounting.** The weight of every attestation is discounted as a function of the graph distance between attester and attested. Signal fabricated within a cluster is structurally cheap.
2. **Community damping.** Dense communities whose internal edge mass exceeds their external evidence by a configured proportion are damped as a group (detection and enforcement in fixed-point arithmetic, deterministic).
3. **Vouch bonds and quotas** (§4.2) bound the fan-out of fabricated identities financeable by an attacker with a given reputation budget.
4. **Judicial layer.** Collusion proven before a jury (§8) is sanctioned across all four domains simultaneously, the sole case of cross-domain sanction (§6).

6. Sanctions and the Liability Cascade

Trust failure propagates liability with three properties, each validated against the specific attack it targets:

- **Capped proportionality.** The sponsor's loss from a sponsored member's fraud is bounded per incident. This closes the **vouch-kamikaze** attack: using a massive fraud from a disposable identity to annihilate an honest sponsor.
- **Probation window.** Fraud committed immediately after receiving a vouch does not propagate a cascade. This closes the **vouch-bomb** attack: soliciting a vouch and detonating it within the same window.
- **Domain locality.** The sanction lands in the domain of the offense; the sole cross-domain exception is structural betrayal (proven Sybil operation, proven collusion, attack on the network).

Sanctions are graduated (public mark, suspension, expulsion), prioritize restitution, and are reversible through the same process that imposes them.

7. Consensus

7.1 Architecture

Block production is decoupled from finality. Production uses a conventional round-based mechanism; **finality** is granted by a committee drawn by sortition weighted by agg(s) (§4.4), rotated every epoch. A block is final once it accumulates signed votes representing a supermajority of the sitting committee's reputational weight.

7.2 Votes

Votes are domain-separated sr25519 signatures over (height, hash). At the gossip layer, each vote's signature is verified before it is relayed, so a peer flooding the network with forged votes is not amplified through honest nodes. At the tally, each vote passes the committee-membership check and signature verification, and is deduplicated by (height, voter). Epoch rotation prunes and refreshes committee state at the transition; a boundary race condition (outgoing committee members voting across the edge), discovered in multi-node dropout-and-rejoin testing, was reproduced and closed (validation ledger, rows 4 and 33).

7.3 The Randomness Beacon

Committee and jury sortition is seeded by a **commit-reveal** beacon: participants commit a value, reveal it in the following window, and the combined entropy of the reveals seeds the selection. Two hardenings:

1. **Last-revealer bias.** We propose layering a verifiable delay function [4] over the beacon's output, removing the advantage of deciding whether or not to reveal after having seen the other reveals.
2. **Reputable-weight guard.** The original design froze selection whenever the set of reveals was weak, which allowed a single reputation-less participant — committing and never revealing — to halt the entire network's finality indefinitely. The corrected version defers to the beacon only when the committed set accumulates sufficient reputable weight; otherwise selection falls back to a deterministic backup mechanism. The attack was reproduced in test before the fix and fails after it.

7.4 The Entrenchment Guard

An on-chain guard monitors the concentration of finality power. If any single entity's weight exceeds a configured ceiling, finality is **reversibly frozen** and recovers on its own once concentration re-disperses. The justification is security-theoretic: halting finality is a recoverable liveness failure; finalizing history under a captured committee is an irreversible safety failure. The guard prefers the former.

8. Dispute Resolution

Disputes between members are resolved by sortitioned juries, with the sortition weighted by the judgment domain and three structural constraints:

1. **Interested-party exclusion:** the parties and any identity with measurable graph proximity to the dispute are excluded from the pool.
2. **Pairwise independence:** the jury must satisfy a minimum graph distance between every pair of jurors — a dispute cannot be adjudicated by a cluster.
3. **Supermajority closure:** the verdict is decided by a supermajority of the sortitioned jury; the two preceding protections ensure that this majority is not that of an interested clan. (An additional layer scoring each juror for coherence with the evidence — rewarding or penalizing the vote according to its alignment with the case file, along the lines of a Schelling point [11] — is future work, §12; it is not implemented today.)

Two-hop collusion jury capture was quantified in simulation (8–10 $\times 10^3$ trials per configuration) and closed with depth-2 weight penalties, verified in unit tests and on devnet. The verdict acts solely on the pseudonym and the proven fact; the system holds no information with which to de-pseudonymize, and no procedure generates such information (§12).

9. Economic Model

Dual asset. A reserve asset (SOV; hard cap 54×10^6, indivisible) and a medium of exchange (HLQ; hard cap 540×10^6, finely divisible). Transaction fees are paid in HLQ and validated against the same ledger that collects them — a mismatch between the fee-validation ledger and the collection ledger, found under adversarial stress testing, would have prevented every new account from paying its first fee; it is fixed and covered by regression.

Issuance by measured service. There is no calendar-based issuance. Minting is tied to algorithmically measured participation (block authorship, finality service, verified infrastructure service); if service is not rendered in a given epoch, the corresponding issuance does not occur and is not consumed against the cap (the phantom-mint defect — consuming the cap before verifying recipients — is closed and verified).

Disinflation. Issuance follows a decreasing curve with a half-life of $\approx$ 24 months (per-epoch retention $\rho = 0.999052$), consistent with the half-life of reputational decay: both layers implement the same property — early advantage dilutes.

Separation of rewards. Committee membership (governing) follows reputation and carries no additional reward; infrastructure service (maintaining) is paid in HLQ, is open to any identity, and confers no governance weight. Founding identities receive no recurring issuance: their genesis allocation is a one-time premine accounted against the cap, and since minting is tied to measured service, no service means no issuance (§10).

Per-person drip. Base citizenship carries a small associated flow of HLQ per person, conditioned on proof of unique personhood (§12), not on legal-identity verification.

Open parameters: the exact shape of the issuance curve; λ; the proof-of-service schedule; the internal split of rewards. These will be modeled and published before genesis (§12).

10. Bootstrap Procedure

Bootstrapping a reputation economy is its most delicate security problem: every initial asymmetry is a seed for capture. Genesis rules:

1. **Minimal, public, and expiring seeding.** Genesis seeds a founding cohort labeled as scaffolding, with decay active from block 0 and zero issuance. Verifiable ceremony rule: every seeded identity corresponds to an operating node at genesis.
2. **Protocol-enforced founder dilution.** The founding nodes' advantage dissolves through reputation decay (§4.5): without attested activity, their weight falls below that of the average citizen in little more than one half-life, with no human intervention. The bootstrap key that admits the initial evidence types expires automatically at a fixed block height — by protocol, not by goodwill — a condition implemented and tested on hardware (ledger, row 14). The founding upgrade authority (a multisig, valid only during construction) additionally carries an expiration in the upgrade pallet itself, with a convergence margin

still under review (§12). An explicit sunset of the founding *validator role* tied to an on-chain dispersion metric is future work (§12), not a mechanism wired today: effective dispersion is produced, for now, by decay.

3. **Concentration ceiling** (§7.4), active from block 0, when the network is smallest and most vulnerable.
4. **External entropy.** The genesis protocol anchors the seed to a public Bitcoin block fixed at the ceremony, removing the founders' own ability to grind their genesis. (The launch ceremony has not yet been executed; until then the value is a placeholder.)
5. Bootstrap-window capture and expiration were adversarially swept in simulation before any value existed (ledger, row 10), and capture-free bootstrap was additionally validated on hardware (row 13).

11. Evaluation

Methodology. The project operates under an evidence-over-claim protocol: no property is declared without a run that exhibits it, and audits precede the existence of value. Multi-node behavior is validated on devnets in two independently operated environments; findings are cross-checked against independent adversarial review panels under an anti-hallucination protocol (every claim is verified against a run, never accepted from prose). A **public validation ledger** ties every claim to the run that supports it, the binary hash, and the raw outputs (33 rows as of this version's close).

Selected results.

Property	Method	Result
Reputation engine resistance (Sybil rings, collusion clusters, targeted slashing, whitewashing)	adversarial battery	17/17 scenarios resisted
Consensus safety and liveness under validator dropout and rejoin across epoch boundaries	multi-node devnet, two environments	survival verified; boundary race closed
Two-hop collusion jury capture	simulation, 8–10 $\times 10^3$ trials/configuration	quantified; closed with depth-2 penalty
Finality freeze via beacon (reveal withholding)	test reproduction	reproduced; closed with reputable-weight guard
Vouch-kamikaze and vouch-bomb	adversarial battery	closed (per-incident caps; probation window)
Determinism of reputation computation	cross-recomputation	bit-identical across nodes and architectures
Fee validation/collection mismatch (unpayable new account)	adversarial stress test	found; fixed; regression active

Normative layer. The system's constitutional document was subjected to the same adversarial process (12 flaws found and resolved before sealing), and every mechanism in this paper is traced to the normative article it serves.

12. Limitations and Future Work

We declare the open residuals rather than omit them:

- **Proof of unique personhood.** The unique-personhood gate has no on-chain implementation yet. The reputation wall (§4) makes it non-critical to consensus security; it remains necessary for the per-person drip (§9). An objective, substitutable mechanism, under evaluation.
- **Transport metadata.** Content is end-to-end blind, but network-layer correlation (timing, sizes, addresses) is the real privacy frontier. Under construction: an onion-routed transport layer. Until then, the honest claim is content privacy, not traffic privacy.
- **Key separation.** Authentication and spending currently share a single key with cryptographic domain separation (v1); separation into subkeys is designed and pending.
- **Parameters under calibration.** λ (a sweep is underway with the full adversarial battery), the shape of the issuance curve, and the proof-of-service schedule: these will be fixed with published evidence before genesis, never by decree.
- **Beacon VDF.** Proposed, not implemented; the beacon currently operates with the reputable-weight guard and deterministic fallback.
- **Finality vote diffusion.** An internal audit found that the diffusion (gossip) layer relayed finality votes without verifying their sr25519 signature before propagating them, which allowed an external peer to amplify forged votes across the network (validity was still checked at the tally, so finality safety was unaffected; it only opened an amplification vector). Fixed: the signature is now verified before relaying (§7.2); pending hardware validation on the relaunch binary.
- **Committee entrenchment guard.** The guard's threshold uses an absolute floor of reputable accounts, not a fraction of the reputable set; this leaves a narrow, self-healing window during bootstrap, when the set is small. An acknowledged, low-severity residual.
- **Dependencies.** The underlying chain framework carries known advisories in transitive dependencies, tracked; the analysis of which ones reach the live node's surface is ongoing.

13. Conclusion

Woven Trust replaces the economic anchor of open consensus — energy or capital, both purchasable — with non-transferable reputational history, computed deterministically from objective evidence and defended in depth against Sybil attacks and collusion. The combination of multi-domain concave aggregation, vouch bonds with a capped cascade, beacon-seeded sortition hardened against grinding, and an entrenchment guard with reversible halt yields a system in which the cost of capturing finality is time spent in verified conduct within the very system one seeks to capture. The published adversarial evaluation — with its declared residuals — supports, we believe, the central claim: a network in which authority is not for sale is implementable with today's tools, and its security is measurable.

References

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The source code, the validation ledger, and the evidence bundles are public and independently verifiable.